Broken by Default: A Formal Verification Study of Security Vulnerabilities in AI-Generated Code

Empirical Analysis of 250 Code Artifacts Across Five Large Language Models Using Z3 SMT Solver, Ablation Studies, and Static Tool Comparison

Dominik Blain

CEO & Co-Founder, QreativeLab
Gatineau, QC, Canada
dominik@qreativelab.io

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Abstract

We present *Broken by Default*, a systematic empirical study of security vulnerabilities in code generated by five major large language models (LLMs): GPT-4o, Claude 3.5 Sonnet, Gemini 2.5 Flash, Mistral Large, and Llama 3.3 70B. Using a benchmark of 50 security-critical prompts spanning five Common Weakness Enumeration (CWE) classes—buffer size calculation (CWE-131), integer overflow (CWE-190), signed/unsigned confusion (CWE-195), weak password hashing (CWE-916), and injection vulnerabilities (CWE-89)—we collected 250 code artifacts and subjected each to formal verification via the Z3 SMT solver within the COBALT analysis pipeline.

Our findings reveal a mean vulnerability rate of 64.8% across all models. Four of five models received a failing grade (F), with Llama 3.3 70B exhibiting the highest vulnerability rate at 68% and Gemini 2.5 Flash performing best at 60% (grade D). A total of **90 findings** were formally proven exploitable via Z3 satisfiability witnesses, and 6 of 7 selected vulnerabilities were confirmed with real memory corruption crashes using GCC AddressSanitizer (ASAN). We further conduct three additional experiments: (1) a secure prompt ablation showing that explicit security instructions reduce the mean vulnerability rate by only 4 percentage points (to 60.8%), leaving four of five models at grade F; (2) a static tool comparison showing that seven industry-standard tools (Semgrep, Bandit, Cppcheck, Clang SA, FlawFinder, and CodeQL security-extended) together detect fewer than 7.6% of all artifacts, and miss 97.8% of Z3-proven vulnerabilities, demonstrating a structural detection gap; and (3) a generation–review asymmetry experiment showing that models identify their own Z3-proven vulnerabilities 78.7% of the time in review mode (70 of 89 artifacts; per-model range: 67%–100%) —yet generate them anyway. Our results suggest that current LLMs produce security-deficient code by default, that security instructions are insufficient mitigations, and that formal verification is the only approach capable of establishing ground-truth exploitability at scale.

1 Introduction

The rapid adoption of AI-assisted coding tools has fundamentally changed software development workflows. GitHub Copilot reported over 1.8 million paid subscribers by 2024 [1]. ChatGPT, Claude, and Gemini are routinely used to generate production code across security-sensitive domains including web backends, embedded systems, and cryptographic libraries.

Yet the security properties of AI-generated code remain understudied. Prior work [2–4] has identified categories of vulnerabilities in LLM outputs, but most rely on static pattern matching or manual inspection—techniques that cannot definitively establish exploitability.

Our contribution is methodological: we apply formal verification via the Z3 Satisfiability Modulo Theories (SMT) solver to establish ground-truth exploitability of AI-generated code vulnerabilities. Where a Z3 witness exists, the vulnerability is not a potential issue—it is a proven one, with a concrete input that triggers the fault condition. We further validate selected findings with compiler-instrumented runtime crashes using GCC AddressSanitizer.

Our findings reveal a mean vulnerability rate of 64.8% across all models. Four of five models received a failing grade (F), with Llama 3.3 70B exhibiting the highest

vulnerability rate at 68% and Gemini 2.5 Flash performing best at 60% (grade D). A total of 90 findings were formally proven exploitable via Z3 satisfiability witnesses, and 6 of 7 selected vulnerabilities were confirmed with real memory corruption crashes using GCC AddressSanitizer (ASAN). We further conduct three additional experiments: (1) a secure prompt ablation showing that explicit security instructions reduce the mean vulnerability rate by only 4 percentage points (to 60.8%), leaving four of five models at grade F; (2) a static tool comparison showing that all seven industry tools combined flag only 7.6% of artifacts, missing 97.8% of formally Z3-proven vulnerabilities, demonstrating an 8.5× detection gap against COBALT; and (3) a generation–review asymmetry experiment showing that models identify their own Z3-proven vulnerabilities 78.7% of the time in review mode (70/89; per-model range: 67%–100%)—yet generate them anyway. Our results suggest that current LLMs produce security-deficient code by default, that security instructions are insufficient mitigations, and that formal verification is the only approach capable of establishing ground-truth exploitability at scale.

1.1 Research Questions

- **RQ1:** At what rate do leading LLMs produce vulnerable code when prompted for security-sensitive tasks?
- **RQ2:** How do vulnerability rates and severity distributions differ across models?
- **RQ3:** Can Z3 SMT witnesses be operationalized into real runtime exploits?
- **RQ4:** Do explicit security instructions in the system prompt reduce vulnerability rates?
- **RQ5:** How does COBALT's detection rate compare to industry-standard static analysis tools?
- **RQ6:** Do models possess the knowledge to detect their own generated vulnerabilities?

1.2 Scope and Limitations

This study targets a bounded set of 50 prompt templates, each designed to elicit code in a specific CWE category. Results reflect model behavior on this prompt corpus as of April 2026 and should not be extrapolated to all possible LLM usage patterns. Models were queried via their production APIs at temperature 0 (deterministic mode). Prompt templates are released alongside this paper.

2 Related Work

Pearce et al. (2022) [2] evaluated GitHub Copilot on 89 scenarios across 18 CWEs and found 40% of suggestions contained vulnerabilities. Their analysis relied on CWE pattern matching without formal exploitability proofs.

Sandoval et al. (2023) [3] studied security of Copilot suggestions for C code, finding that users who accepted suggestions without review introduced more vulnerabilities than those who wrote code from scratch.

Tony et al. (2023) [4] introduced LLMSecEval, a benchmark of 150 natural-language prompts and evaluated 5 LLMs, finding high rates of insecure code generation particularly for injection and buffer-related vulnerabilities.

Perry et al. (2023) [5] conducted a user study where participants using AI assistants wrote significantly more security vulnerabilities than those who did not, despite expressing higher confidence in their code.

Meta CyberSecEval (2024) [11] benchmarks LLMs on generating exploit code and assisting cyberattacks. Their focus is adversarial capability—can the model help an attacker? Our focus is orthogonal: does the model introduce vulnerabilities in everyday code generation, without any adversarial intent?

SecurityEval (2022) [12] provides 130 Python prompts derived from CWE patterns and evaluates GitHub Copilot. Like Pearce et al., evaluation relies on pattern-based analysis. No formal exploitability proof is provided.

SVEN (2023) [13] trains a security vector to steer code generation toward secure outputs. Their setup assumes the vulnerability exists; ours measures baseline generation without steering intervention.

Our work differs in four key dimensions: (1) we apply formal Z3 SMT verification rather than pattern matching—producing mathematical proof of exploitability with concrete witness values; (2) we evaluate five current frontier models simultaneously across identical prompts; (3) we provide runtime exploitability confirmation via ASAN; and (4) we quantify the detection gap against seven industry tools, including heavyweight C analyzers and CodeQL, demonstrating that 97.8% of formally proven vulnerabilities are structurally invisible to existing tooling.

3 Methodology

3.1 Benchmark Design

We constructed a benchmark of 50 natural-language prompts distributed across five CWE categories (10 prompts each):

- **MEM** (CWE-131/190): Buffer allocation, array indexing, dynamic memory management in C/C++
- **INT** (CWE-190/195): Integer arithmetic, type casting, signed/unsigned arithmetic
- **AUTH** (CWE-916): Password hashing, authentication token generation, session management
- **CRYPTO** (CWE-327/330): Key generation, random number usage, cipher selection
- **INP** (CWE-89/22/78): SQL construction, file path handling, shell command composition

Prompts were designed to be representative of real developer tasks—not adversarial jailbreaks. Example prompt (MEM-01):

> *"Write a C function that reads n integers from a file descriptor and returns them as a dynamically allocated array."*

3.2 Model Selection and API Configuration

Five models were evaluated, selected to represent the frontier of publicly accessible LLMs as of Q1 2026:

Model	API Identifier
GPT-4o	`gpt-4o`
Claude 3.5 Sonnet	`claude-3-5-sonnet-20241022`
Gemini 2.5 Flash	`gemini-2.5-flash`
Mistral Large	`mistral-large-latest`
Llama 3.3 70B	`llama-3.3-70b-versatile`

All models were queried at temperature 0 to ensure reproducibility. System prompts instructed models to produce complete, runnable code without additional commentary, maximizing the proportion of testable output.

3.3 COBALT Analysis Pipeline

Generated code artifacts were analyzed using the COBALT static analysis engine, which combines:

1. **CWE Pattern Extraction**: Regex and AST-level patterns identify candidate vulnerability sites (e.g., `malloc(n * sizeof(T))` without overflow guard)
2. **Z3 SMT Encoding**: Vulnerability conditions are encoded as Z3 formulas. For integer overflow: given n as a free variable of type `BitVec(32)`, check satisfiability of $n \times sizeof(T) < n$ (overflow condition under unsigned 32-bit semantics)
3. **Witness Extraction**: When Z3 returns SAT, the concrete witness value (e.g., $n = 2^{30} + 1$) is extracted and classified as the exploit input

Findings were classified as:

- **Z3 SAT**: Formally proven exploitable via SMT witness
- **PATTERN MATCH**: Structural vulnerability identified, Z3 encoding pending
- **CLEAN**: No vulnerability detected

Severity was assigned using CVSS v3 base score criteria: CRITICAL (≥9.0), HIGH (7.0–8.9), MEDIUM (4.0–6.9).

3.4 Runtime Exploitability Validation

To counter the objection that pattern-based detection inflates vulnerability counts, we selected 7 representative findings and constructed C and Python proof-of-concept (PoC) harnesses. C PoCs were compiled with:

```
gcc -fsanitize=address,undefined -g -O1 \
  -o poc poc.c && ./poc
```

Z3-extracted witness values were used as concrete inputs. We captured ASAN output to confirm runtime faults.

4 Results

4.1 RQ1: Overall Vulnerability Rates

Table 1 presents aggregate results across all 50 prompts per model.

Table 1: Benchmark Leaderboard — 50 Prompts per Model

Model	Vuln	CRIT	HIGH	Z3	Grd
Llama 3.3 70B	68%	22	7	17	F
GPT-4o	66%	21	7	18	F
Mistral Large	66%	21	6	18	F
Claude 3.5 Sonnet	64%	21	7	19	F
Gemini 2.5 Flash	60%	20	6	18	D
Mean	64.8%	21	6.6	18	—

The mean vulnerability rate of 64.8% is alarmingly high. Even the best-performing model (Gemini 2.5

Flash at 60%) produced vulnerable code in 30 of 50 test cases. CRITICAL-severity findings dominated: an average of 21 CRITICAL findings per model, representing vulnerabilities directly exploitable with attacker-controlled inputs.

4.2 RQ2: Vulnerability Distribution by CWE

Table 2 illustrates that memory safety vulnerabilities (CWE-131, CWE-190) accounted for the largest share of findings across all models, consistent with known challenges in generating safe C/C++ code.

Table 2: Vulnerability Rate by Category (Mean Across Models)

Category	CWE(s)	Mean Rate
Memory Allocation	131, 190	78%
Integer Arithmetic	190, 195	72%
Authentication	916	60%
Cryptography	327, 330	58%
Input Handling	89, 22, 78	56%

Memory allocation prompts produced the highest vulnerability rate (78%), driven by consistent failure to guard against integer overflow in malloc size computations. A representative pattern found across all five models:

Listing 1: Typical vulnerable pattern (all 5 models)

```
// CWE-190: No overflow guard
int* buf = malloc(n * sizeof(int));
```

The safe pattern requires an explicit overflow check:

Listing 2: Correct pattern

```
if (n > SIZE_MAX / sizeof(int)) return
    NULL;
int* buf = malloc(n * sizeof(int));
```

None of the five models consistently generated the safe pattern across all MEM prompts.

4.3 RQ3: Runtime Exploitability Confirmation

Of 7 selected vulnerabilities subjected to PoC harness testing, 6 of 7 produced confirmed runtime faults. Table 3 summarizes the results.

* Python 3.12 raises `ValueError` on path traversal. Vulnerable pattern present; blocked at runtime, not at generation.

Representative ASAN output for MEM-01-A (Llama, CWE-131):

Table 3: PoC Exploit Confirmation Results

ID	Model	Fault Type	Result
MEM-01-A	Llama	heap-buf-overflow	CONFIRMED
MEM-01-B	GPT-4o	heap-buf-overflow	CONFIRMED
MEM-03	Llama	alloc-size-too-big	CONFIRMED
MEM-06	GPT-4o	OOB read	CONFIRMED
AUTH-03	Llama	SHA-256 crack	CONFIRMED
INP-01	Mistral	SQL injection	CONFIRMED
INP-06	GPT-4o	Zip Slip (path trav.)	BLOCKED*

Listing 3: ASAN output – MEM-01-A heap overflow

```
==AddressSanitizer: heap-buffer-overflow
WRITE of size 4 at 0x...
#0 poc_main (poc+0x...)
#1 main (poc+0x...)
shadow bytes around the buggy address:
0x...: fa fa fa fa fa fa fa fa
0x...: 00 00 00 00 00[fa]fa fa
SUMMARY: AddressSanitizer: heap-buffer-
    overflow
```

For AUTH-03 (SHA-256 password cracking), a 6-character lowercase password was recovered from its hash in 0.01ms using a precomputed lookup, demonstrating that CWE-916 (insufficient password hashing) is not merely theoretical.

SQL injection in INP-01 produced a working query that extracted all records including a synthetic credit card number embedded in the test database, confirming complete data exfiltration.

4.4 RQ4: Secure Prompt Ablation

We re-ran all 50 prompts for all five models with a security-explicit system prompt instructing models to apply security best practices, guard against integer overflow, and produce production-ready code. Table 4 shows the results.

Table 4: Baseline vs. Secure Prompt Ablation

Model	Base	Secure	Δ	Grade
Llama 3.3 70B	68%	70%	+2%	F
GPT-4o	66%	58%	−8%	F
Gemini 2.5 Flash	60%	60%	0%	F
Claude 3.5 Sonnet	64%	62%	−2%	F
Mistral Large	66%	54%	−12%	D
Mean	64.8%	60.8%	−4%	—

Explicit security instructions reduced the mean vulnerability rate by only 4 percentage points—from 64.8% to 60.8%. Four of five models remained at

grade F. Llama 3.3 70B performed worse under the secure prompt (+2%). The improvement was category-dependent: authentication and cryptography showed modest gains, while memory allocation vulnerabilities were essentially unchanged across all models—suggesting that security instructions do not override low-level memory management patterns learned from training data.

4.5 RQ5: Static Tool Comparison

We conducted three tool comparison experiments. First, we ran Semgrep (all rulesets) and Bandit across all 250 artifacts. Second, we ran three heavyweight C static analyzers—Cppcheck 2.13, Clang Static Analyzer, and FlawFinder 2.0—against all 87 C artifacts. Third, we ran CodeQL (security-extended query suite, version 2.x) against the 90 Z3 SAT artifacts to assess detection of formally proven findings. Table 5 summarizes all three experiments.

Of the 162 COBALT findings, 90 are Z3 SAT—formally proven with concrete arithmetic witnesses—and 72 are PATTERN MATCH, flagged by COBALT's AST layer without Z3 confirmation.

Pattern tools (Experiment A). All 19 tool-caught artifacts overlap with COBALT's PATTERN MATCH tier or dangerous-function patterns (`strcat`, `rand`), never with the integer overflow class. Of the 90 Z3 SAT findings, tools caught only 2—both via Semgrep's `insecure-use-strcat-fn` rule applied to MEM-02 (CWE-190) code that also contained `strcat`, not because any tool detected the integer overflow in the length computation. 97.8% (88/90) of formally Z3-proven vulnerabilities are invisible to all industry tools combined.

Heavyweight C analyzers (Experiment B). Cppcheck and Clang Static Analyzer detected zero of 80 confirmed C vulnerabilities. FlawFinder flagged 4 artifacts: 2 for `strcat` (MEM-02, CWE-190) and 2 for weak randomness (`rand`, CRYPTO-08, CWE-338)—no detection of integer overflow in allocation arithmetic. Combined, heavyweight analyzers missed 96.9% of COBALT findings on C.

This is not a configuration issue. We ran Cppcheck with `--enable=all --inconclusive --check-level=exhaustive`; only 113 of 592 available checkers activate on standard C11 code. Detecting `malloc(n * sizeof(T))` as an integer overflow requires arithmetic semantic reasoning under 32-bit modular arithmetic—which Z3 provides and which neither pattern matching nor path-sensitive analysis can perform without concrete value constraints or explicit taint propagation from attacker-controlled inputs.

CodeQL v2.25.1 (Experiment C). CodeQL (security-extended query suite, v2.25.1, native x86-64 on GitHub Actions) was run against all 90 formally Z3-proven findings: 68 C artifacts and 22 Python artifacts. CodeQL detected **0 of 68 C artifacts** and **0 of 22 Python artifacts**—a 100% miss rate across both languages (0/90 total). This result is consistent with Experiments A and B: integer overflow in allocation arithmetic (`malloc(n * sizeof(T))`) is structurally undetectable by dataflow and taint-tracking analysis without concrete arithmetic reasoning. CodeQL's integer overflow queries require explicit source-to-sink taint paths or known unsafe patterns—a property that Z3's bit-vector arithmetic encodes directly but that CodeQL's query model does not generalize to.

4.6 RQ6: The Generation–Review Asymmetry

A plausible counter-argument to our findings is that models generate vulnerable code simply because they are not asked to think about security. To test this, we conducted a self-review experiment: we fed each model's Z3-proven vulnerable code back to the same model and asked it to review the code for security vulnerabilities.

We fed all 89 valid Z3-proven artifacts back to their generating model and asked it to review the code for security vulnerabilities. (One artifact was excluded: INP-06, where the Python 3.12 runtime blocked the exploit before execution.) Of 89 artifacts reviewed, **70 of 89 (78.7%)** were correctly identified as vulnerable by the generating model. Table 6 presents per-model results.

This finding is more damning than a simple false-negative result. It reveals an *observed generation–review asymmetry*: models possess the knowledge to identify these vulnerability classes when in review mode, yet consistently fail to apply that knowledge during code generation. The problem is not a lack of security knowledge—it is a failure of spontaneous application at a 21.3% false-negative rate across the full artifact set. This suggests that RLHF or instruction fine-tuning aimed at security-conscious code review does not transfer reliably to the code generation task.

5 Discussion

5.1 Why Do Models Fail Consistently?

The consistency of failures—particularly in memory allocation—suggests a training data effect. C code on the internet overwhelmingly lacks overflow guards for

Table 5: Detection Rate: COBALT vs. Seven Industry Tools

Tool	Det. Rate	C / Py	Note
COBALT Z3	162/250 (64.8%)	80/87 · 82/163	
Z3 formally proven	90/250 (36.0%)		Pattern-based tools (Exp. A):
Semgrep (all rulesets)	16/250 (6.4%)	2/87 · 14/163	
Bandit (medium+ severity)	5/250 (2.0%)	— · 5/163	
Combined A	19/250 (7.6%)	—	
Heavyweight C analyzers (Exp. B, C only):			
Cppcheck 2.13	0/87 (0.0%)	0/87	
Clang Static Analyzer	0/87 (0.0%)	0/87	
FlawFinder 2.0 (risk≥3)	4/87 (4.6%)	4/87	strcat + rand only
Combined B	4/87 (4.6%)	—	
Z3 SAT formal proof set (Exp. C):			
CodeQL v2.25.1 (security-extended)	0/90 (0.0%)	0/68 · 0/22	100% miss; C + Python
COBALT-only (vs. A)	151/162 (93.2%)	78/80 · 73/82	
COBALT-only (vs. B)	157/162 (96.9%)	76/80 · 81/82	
Z3 SAT-only (vs. A+B+C)	88/90 (97.8%)		formal proofs

Table 6: Self-Review Experiment — Per-Model Results (N=89)

Model	Detected	Rate	FN Rate
Mistral Large	17/17	100%	0%
Llama 3.3 70B	14/17	82%	18%
Gemini 2.5 Flash	14/18	78%	22%
Claude 3.5 Sonnet	13/19	68%	32%
GPT-4o	12/18	67%	33%
Total	70/89	78.7%	21.3%

malloc computations. Models appear to have internalized this pattern as correct. The secure prompt ablation (RQ4) reinforces this hypothesis: even explicit instructions to guard against overflow had no measurable effect on memory allocation prompts.

5.2 Security Instructions Are Insufficient

A mean improvement of 4 percentage points under an explicit security prompt is not a safety margin—it is noise. Developers who add "write secure code" to their prompt prefixes should not interpret this as a meaningful mitigation. Our results suggest that the vulnerability patterns are baked into the model's generation prior, not into its instruction-following surface.

5.3 The COBALT Detection Gap

COBALT's detection advantage—64.8% vs. 6.4% for all Semgrep rulesets and 0% for Cppcheck and Clang SA on C—is not an artifact of overly sensitive rules. Of the 90 Z3 SAT findings (formally proven with concrete witnesses), 97.8% are invisible to all seven tools. The 2 Z3 SAT cases tools did catch (MEM-02, CWE-190) were flagged by Semgrep's `strcat` rule—detecting a dangerous string function in the same code, not the integer overflow that Z3 proved exploitable. CodeQL v2.25.1, despite its reputation as a heavyweight semantic analyzer, likewise detected zero of 90 Z3-proven findings (0/68 C, 0/22 Python). No tool, across any configuration, detected a single integer overflow in malloc size computations. The detection gap reflects a structural limitation: syntactic tools match patterns in code; path-sensitive tools track values along known execution paths. Neither approach can determine that `malloc(n * sizeof(T))` is dangerous unless n is provably bounded—a property that requires arithmetic reasoning across the full domain of integer inputs, which Z3's bit-vector arithmetic encodes directly.

5.4 Implications for Developers

1. Treat AI-generated C/C++ as unreviewed code requiring explicit security audit
2. Apply `-fsanitize=address,undefined` to all AI-generated systems code
3. Do not rely on security prompt prefixes as a

substitute for review

4. Do not rely on Semgrep, Bandit, Cppcheck, Clang SA, FlawFinder, or CodeQL alone to catch AI-generated integer overflow in allocation arithmetic
5. Use formal verification or human review for allocation arithmetic

5.5 Limitations

Our benchmark of 50 prompts covers five CWE classes; other vulnerability categories may show different rates. All models were tested at temperature 0; higher temperatures may yield different distributions. Of 162 COBALT findings, 90 are Z3 SAT (formally proven) and 72 are PATTERN MATCH (AST-layer detections without Z3 confirmation); the latter class may include false positives not evaluated here. Self-review was conducted on all 89 valid Z3-proven artifacts (excluding one runtime-blocked case); per-model false-negative rates ranged from 0% to 33%. The secure prompt ablation showed a 2-point increase for one model (Llama), which on a 50-prompt sample represents one additional artifact and may be within noise. We did not test multi-turn interactions or retrieval-augmented generation contexts.

5.6 Ethical Considerations

All artifacts were generated under controlled conditions. PoC exploits targeted only locally generated test programs. No production systems were accessed. All prompts, results, and scripts are released to support reproducibility and independent replication. The full dataset of 250 artifacts with Z3 labels is publicly available at https://github.com/dom-omg/bbd-dataset.

6 Conclusion

We presented a multi-experiment formal verification study of AI-generated code security, evaluating five frontier LLMs across 250 code artifacts and three experimental conditions. Our findings converge on a single conclusion: AI coding assistants are broken by default.

The baseline vulnerability rate of 64.8% drops to only 60.8% under explicit security instructions—a 4-point reduction that leaves four of five models at grade F. Seven industry-standard tools (Semgrep, Bandit, Cppcheck, Clang SA, FlawFinder, and CodeQL security-extended) combined flag only 7.6% of all artifacts, and miss 97.8% of formally Z3-proven vulnerabilities—not because of misconfiguration, but because integer overflow in allocation arithmetic is structurally undetectable by pattern and path-sensitive analysis. Models that generate vulnerable code correctly identify those vulnerabilities in review mode 78.7% of the time—proving the problem is not ignorance but a structural failure in how code generation and security knowledge interact.

Formal verification via Z3 SMT solving is not a niche research tool. It is the only approach that can establish ground-truth exploitability of the vulnerability patterns that dominate AI-generated code. This methodology should be a mandatory component of any AI code review pipeline operating on memory-sensitive or security-critical targets.

Future work will extend the benchmark to 200+ prompts, add multi-language targets (Go, Rust, JavaScript), and evaluate the effect of targeted security fine-tuning on these specific CWE classes.

Reproducibility

Benchmark prompts, raw results (JSON), analysis scripts, and PoC harnesses are available at: https://github.com/dom-omg/broken-by-default

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